Course Assessment

The same paper is used before and after the course, so that we can measure how much the teaching moved you. **It is not graded and it does not affect your marks.**

PLEASE READ

Answer honestly. If you do not know something, choose the answer you think is most likely and move on — **do not guess wildly and do not look anything up**. An honest wrong answer before the course is far more useful to us than a lucky right one, because the gap between the two papers is the entire measurement.

About 10 minutes. 20 multiple-choice questions and 5 statements about how confident you feel.

NAME OR STUDENT ID

DATE

TICK ONE

- ▢ BEFORE the course
- ▢ AFTER the course

1 Multiple choice

Tick one answer per question.

1. **One Sentinel-2 pixel covers an area of about:**
   - ▢ **A** 1 × 1 metre
   - ▢ **B** 10 × 10 metres
   - ▢ **C** 100 × 100 metres
   - ▢ **D** 1 × 1 kilometre

2. **A satellite "band" is:**
   - ▢ **A** The orbit the satellite follows
   - ▢ **B** The measurement of one specific wavelength of light
   - ▢ **C** A group of pixels of the same colour
   - ▢ **D** A strip of land the satellite photographs

3. **Healthy green leaves are much brighter than soil or water in which part of the spectrum?**
   - ▢ **A** Red
   - ▢ **B** Thermal
   - ▢ **C** Blue
   - ▢ **D** Near-infrared

4. **Why do we build a median composite from many dates instead of using one clear image?**
   - ▢ **A** To increase the spatial resolution
   - ▢ **B** It makes the file smaller
   - ▢ **C** Because in a monsoon climate there may be no single cloud-free date
   - ▢ **D** Because satellites only pass over once a year

5. In a false-colour image using near-infrared, healthy forest appears:

- ▢ **A** Dark blue
- ▢ **B** White
- ▢ **C** Black
- ▢ **D** Bright red

6. When you run an Earth Engine analysis, where does the computation happen?

- ▢ **A** On your laptop, after downloading the images
- ▢ **B** In your web browser
- ▢ **C** On Google servers, next to the stored data
- ▢ **D** On the satellite itself

7. What does the `scale` argument to reduceRegion control?

- ▢ **A** How many decimal places are returned
- ▢ **B** The pixel size used for the calculation, in metres
- ▢ **C** The colour range of the display
- ▢ **D** The size of the output map on screen

8. An ImageCollection differs from an Image in that it:

- ▢ **A** Contains many images, usually across time
- ▢ **B** Can only hold vector data
- ▢ **C** Is stored at higher resolution
- ▢ **D** Has more bands

9. Your free Earth Engine account gives you a monthly computing budget. The safest habit is to:

- ▢ **A** Study one province at a time and set an explicit scale
- ▢ **B** Export everything to Drive as GeoTIFF
- ▢ **C** Always use the finest scale available
- ▢ **D** Run every analysis twice to check it

10. NDVI is calculated as:

- ▢ **A** NIR / Red
- ▢ **B** (NIR - Red) / (NIR + Red)
- ▢ **C** (Red - NIR) / (Red + NIR)
- ▢ **D** (NIR - SWIR) / (NIR + SWIR)

11. Your NDVI map has a maximum value of 3,412. What went wrong?

- ▢ **A** The reflectance values were not divided by 10,000
- ▢ **B** The scale was set too coarse
- ▢ **C** The forest is unusually healthy
- ▢ **D** The cloud mask removed too many pixels

12. A dense, healthy tropical forest would typically have an NDVI of about:

- ▢ **A** -0.5
- ▢ **B** 0.1
- ▢ **C** 0.4
- ▢ **D** 0.8

13. You cannot separate evergreen from deciduous forest in one image because:

▢ **A** Both look the same when in leaf — the difference is seasonal change
▢ **B** Satellites cannot detect species
▢ **C** The pixels are too large
▢ **D** Deciduous forest is always cloudy

14. Why must classification accuracy be measured on points the model has never seen?

▢ **A** Because training points are always wrong
▢ **B** It does not matter which points you use
▢ **C** To save computing time
▢ **D** Because the model has partly memorised its training points, which inflates the score

15. A colleague reports 99.4% overall accuracy on a four-class forest map. You should:

▢ **A** Congratulate them, that is excellent
▢ **B** Ask them to add more classes
▢ **C** Be suspicious — most likely the test data leaked into training
▢ **D** Assume the satellite was unusually clear that day

16. To convert aboveground biomass (Mg/ha) to carbon, you multiply by approximately:

▢ **A** 0.25
▢ **B** 0.47
▢ **C** 1.0
▢ **D** 3.67

17. A dataset reports its band in Mg C/ha. Before summing it, you should:

▢ **A** Use it as-is — it is already carbon
▢ **B** Convert it to Kelvin
▢ **C** Multiply by 0.47 to get carbon
▢ **D** Divide by 0.47

18. To answer "how many hectares burned last season?", the right dataset is:

▢ **A** FIRMS active fire detections
▢ **B** CHIRPS rainfall
▢ **C** Sentinel-2 true colour
▢ **D** MODIS MCD64A1 burned area

19. Applied across a whole province in northern Thailand, dNBR alone reports far too much burned area because:

▢ **A** Clouds are mistaken for smoke
▢ **B** The satellite resolution is too coarse
▢ **C** Deciduous forest dropping its leaves looks spectrally like burnt ground
▢ **D** dNBR only works on conifers

20. ERA5 gives you temperatures around 298. To use them you must:

▢ **A** Multiply by 0.47
▢ **B** Nothing — those are correct in Celsius
▢ **C** Subtract 273.15 to convert Kelvin to Celsius
▢ **D** Divide by 10

2 How confident do you feel?

There are no right answers here. Tick the box that matches how you feel right now.

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1. I could produce a forest cover map for a new area on my own.	☐	☐	☐	☐	☐
2. I can tell whether a number produced by satellite analysis is plausible.	☐	☐	☐	☐	☐
3. I know how to describe a task to an AI assistant so it writes correct code.	☐	☐	☐	☐	☐
4. I know what to do when code an AI gave me produces an error.	☐	☐	☐	☐	☐
5. I could explain to a colleague why two forest maps of the same place disagree.	☐	☐	☐	☐	☐

THANK YOU

If this is the paper you are filling in **before** the course, you are expected to find most of it hard. That is the point — it is measuring the course, not you.